

where the superscript h denotes the finite element subspace. $\mathcal{M}^h$, ℓ^h and $\mathcal{P}^h$ are the finite-element subspaces of the function space $\mathcal{M} \equiv \{\mathbf{u} \mid u_i \in H^1(\Omega); u_i = g_i \text{ on } \Gamma_g\}$, $\ell \equiv \{\mathbf{w} \mid w_i \in H^1(\Omega); w_i = 0 \text{ on } \Gamma_g\}$ and $\mathcal{P} \subseteq L^2(\Omega)$, respectively. The third term is the stabilized term to address the volumetric locking induced by the incompressibility where NE is the total number of elements in the discretized domain. $\tau = \frac{\alpha h^2}{2\mu}$ represents the stabilization factor, where h is the characteristic element length. α is set to 1.0 [30].

Topological description of nonhomogeneous elastic property distribution

To describe the topology of the nonhomogeneous elastic property distribution, we introduce the B-spline function to express a curve:

$$\mathbf{C}(t) = \sum_{i=0}^{n+1} N_{i,p}(t) \mathbf{P}_i, \quad (a \leq t \leq b) \quad (5)$$

where $N_{i,p}(t)$ is the p -th order B-spline basis function generated by a knot vector $\{t_0, t_1, \dots, t_{n+2+p}\}$. $\mathbf{P}_i$ ($i = 0, \dots, n+1$) is the i -th control point.

The function $N_{i,p}(t)$ can be determined by Eq. (6):

$$\begin{cases} N_{i,0}(t) = \begin{cases} 1, & \text{if } t_i \leq t \leq t_{i+1} \\ 0, & \text{otherwise} \end{cases} & (i = 0, \dots, n+1; p = 0) \\ N_{i,p}(t) = \frac{t - t_i}{t_{i+p} - t_i} N_{i,p-1}(t) + \frac{t_{i+p+1} - t}{t_{i+p+1} - t_{i+1}} N_{i+1,p-1}(t), & (i = 0, \dots, n+1; p \geq 1) \end{cases} \quad (6)$$

To capture various types of nonhomogeneous elastic property distributions, we employ B-splines to approximate both open and closed curves, as depicted in Fig. 1. The open curve is utilized to represent the interface shape between adjacent layers in a layered composite, while the closed curve represents the shape of inclusions embedded within a soft solid. To avoid the self-intersection of the closed B-spline curve, we define the control points of the closed B-spline curve in Eq. (7):

$$\mathbf{P}_i = \begin{cases} \left(x_o + d_i \sin\left(i\theta - \frac{\theta}{2}\right), y_o + d_i \cos\left(i\theta - \frac{\theta}{2}\right) \right), & \left(\theta = \frac{2\pi}{n}; i = 1, \dots, n \right) \\ \mathbf{P}_0 = \mathbf{P}_{n+1} = \frac{\mathbf{P}_1 + \mathbf{P}_n}{2} \end{cases} \quad (7)$$

where (x_o, y_o) is the location of the center of the closed B-spline curve, d_i is the distance between the i -th control point and the center of the closed curve.

With the curve expressed by the B-spline function, we can define each nonhomogeneous region in the problem domain by

[Figure 1 — placeholder]

- (a) **Open curve:** eight control points $\mathbf{P}_0, \mathbf{P}_1, \dots, \mathbf{P}_7$ connected by dashed blue segments, with a smooth orange B-spline curve passing through the polygon.
- (b) **Closed curve:** nine control points $\mathbf{P}_0 = \mathbf{P}_9, \mathbf{P}_1, \dots, \mathbf{P}_8$ distributed around a center (x_o, y_o) , with radial distances $d_1, d_2, \dots, d_8$ and angular spacing $\theta = 2\pi/n$. The closed B-spline curve (orange) is inscribed in the dashed control polygon, with $\mathbf{P}_0 = \mathbf{P}_9 = (\mathbf{P}_1 + \mathbf{P}_8)/2$.

Fig. 1. Two examples of the curve expressed by B-spline functions.